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Detailed kinetic modeling of dimethoxymethane. Part II: Experimental and theoretical study of the kinetics and reaction mechanism

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ABSTRACT

In this study (Part II), the oxidation of dimethoxymethane (DMM) is investigated and a detailed chemical reaction model developed for a comprehensive description of both high- and low-temperature oxidation processes. The sub-mechanism of DMM is implemented using AramcoMech2.0 as the base mechanism. Rate coefficients are based on analogies with those for dimethyl ether, diethyl ether, and *n*-pentane oxidation. Furthermore, theoretical studies from recent works are also included in the present model and new calculations for the dissociation kinetics of $\dot{Q}OOH$ radicals have been carried out at the CCSD(T)/CBS(aug-cc-pVXZ; X = D, T) // B2PLYP-D3/6-311++G(d,p) level of theory. For validation, new ignition delay time experiments have been performed in a shock tube (ST), a rapid compression machine (RCM), and in a laminar flow reactor covering a wide range of conditions ($p = 1\text{--}40$ bar, $T = 590\text{--}1215$ K, $\varphi = 1$). In addition, the kinetic model is validated against laminar burning velocities, jet-stirred reactor, plug flow reactor and further ST and RCM experimental datasets from the literature. Pathway and sensitivity analyses were used to identify critical reaction pathways in the DMM oxidation mechanism. These show that the reactivity of DMM at intermediate temperatures is controlled by the branching between pathways initiated on the primary or secondary fuel radical. While primary fuel radicals eventually lead to chain branching, secondary fuel radical consumption is controlled by fast β -scission over a wide range of temperatures, which inhibits reactivity.

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1. Introduction

The production of alternative fuels from renewable energy sources is a promising solution to alleviate the twin challenges of resource consumption and emissions. With a view to meeting strict future emission regulations, modern combustion engines are becoming increasingly efficient. However, in addition to improvements in engine design, alternative fuels also have the potential to optimize the combustion process. In this regard, oxymethylene ethers (OME_x) have been identified as potential future fuel candidates, which can be used pure or as fuel-blending agents. OME_x as neat fuels or as diesel additives are efficient in reduction of soot emission [1–4]. The suppression of sooting is assumed to be due to their C–O moieties imbedded in their saturated molecular

structure, as a result their lack of carbon-carbon bonds, and their high amount of oxygen mass content [1,5].

Several detailed studies have investigated the influence of OME₁, also called dimethoxymethane (DMM) or methylal, on the combustion and emissions characteristics of compression ignition (CI) engines. The results of these investigations show that DMM can improve the NO_x/PM trade-off [6–8], reduce exhaust CO concentration [9] as well as partially reduce combustion noise [10]. The physical properties of DMM include a low cetane number (CN), a high volatility and a low viscosity and lubricity, hence DMM is usually used as a blend with diesel fuel or with additives [1]. Omari et al. [10] examined the effect of different mixture compositions of DMM in diesel. The screenings indicated an optimum, with a blend ratio of 35% DMM in diesel for the best compromise between soot reduction and the deterioration of fuel properties such as heating value and CN. With an increasing chain length of an OME_x ($x = 3\text{--}5$), diesel-like properties become more prominent and OME_x can be used to substitute diesel fuel [1,11]. Additionally, all

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OME_x with $x \geq 2$ have a higher cetane number compared to diesel. Previous investigations of long-chain OME_x blends in a diesel engine with exhaust gas recirculation (EGR) achieved high efficiency and clean combustion [8,12]. Recent studies have investigated the production process of DMM, with two possible DMM synthesis pathways being distinguished; (i) conventional production via the condensation of methanol and formaldehyde (FA route) [13,14] and (ii) a new catalytic transformation allowing the direct synthesis of DMM from methanol, CO₂ and H₂ (direct route) [15,16]. A comparison of the environmental impacts and the exergy efficiency for both DMM synthesis routes were analyzed in the work of Deutz et al. [17] and show that the exergetic efficiency due to the direct route could be improved from 74% to 86%.

All of these potential applications motivate the study of the fundamental combustion chemistry of dimethoxymethane. DMM is a di-ether or acetal and has the simplest linear molecular structure, namely CH₃O-(CH₂O)_x-CH₃ in the OME_x group. Several previous detailed studies have analyzed the fundamental combustion properties, including the auto-ignition characteristics of DMM.

Daly et al. [18] investigated the oxidation of DMM in a jet-stirred reactor (JSR) and developed a kinetic model to simulate the species profiles measured. Renard et al. [19,20] examined the effect of DMM addition on the structure of a fuel-rich, flat, premixed ethylene/oxygen/argon flame and observed a reduction in the peak mole fraction of soot precursors. Sinha and Thomson [21] also measured the concentration profiles of DMM, major products (CO, CO₂), and the more important intermediate hydrocarbons produced during oxidation in an opposed flow diffusion flame burner. They suggested that the formation of ethylene and acetylene is inhibited due to the lack of C-C bonds and that the oxidation of DMM mainly proceeds via H-atom abstraction and successive β -scission reactions to produce methyl radicals and formaldehyde (FA, CH₂O). Dias et al. [22] studied fuel-lean and fuel-rich premixed DMM/oxygen/argon flames and used these data to develop and validate a new DMM sub-mechanism. Ignition delay times (IDTs) of DMM/oxygen/argon mixtures were investigated at high temperatures in a shock tube (ST) by Zhang et al. [23]. A focus of their investigation was a comparison of experimental and predicted ignition delay times of DMM from the available literature. Hu et al. [24] measured the auto-ignition behavior of DMM and DMM/*n*-heptane blends in a ST at temperatures above 1100 K. These authors also developed a new DMM/*n*-heptane blend mechanism based on various literature sources [25,26] showing the effect of DMM addition on the auto-ignition behavior of *n*-heptane. Gillespie [27] measured IDTs for homogeneous DMM mixtures over a wide range of temperatures at various equivalence ratios using rapid compression machine (RCM), low-, and high-pressure shock tube (LPST, HPST). In addition to the auto-ignition experiments, laminar burning velocities of DMM in air were also reported. The oxidation of DMM was investigated using a quartz flow reactor at various equivalence ratios and pressures of up to 60 bar by Marrodán et al. [28,29].

Very few modeling studies have considered the oxidation of DMM in both the high- and low-temperature regimes. However, He et al. [30] recently developed a chemical kinetic mechanism up to OME₃ for low- and intermediate-temperature combustion. The predictions capture well the ignition behavior of OME₃ measured in an RCM. Furthermore, Vermeire et al. [31] investigated the pyrolysis and low- to intermediate-temperature oxidation pathways of DMM in a JSR and developed a new kinetic model for DMM. Both of the aforementioned studies determined the thermodynamic properties of the main species and the modified Arrhenius parameters for several important reactions for DMM oxidation based on quantum mechanical calculations.

In this study (Part II) we develop a new detailed kinetic model which includes high- and low-temperature pathways for a more

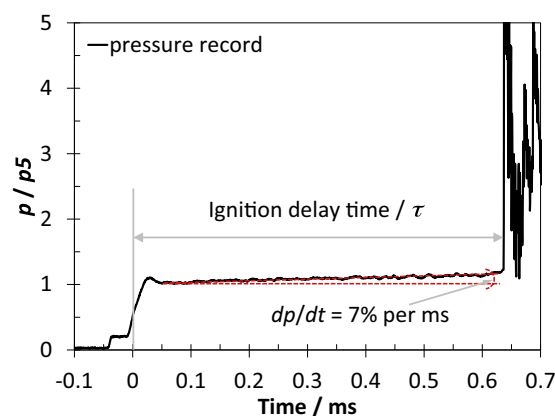


Fig. 1. Pressure trace showing ignition delay time of DMM. Experimental conditions: $p_5 = 19.3$ bar, $T_5 = 919$ K, $\phi = 1.0$ in air and $\tau = 0.64$ ms, pressure rise due to shock attenuation of 7%/ms.

comprehensive description of DMM oxidation at engine-relevant conditions. In Part I of this study, we computed the thermochemistry and kinetics of key species and reactions [32], which are used here. To the best of our knowledge there is no fundamental research of the auto-ignition characteristics of neat gas-phase DMM in air mixtures at engine-relevant conditions. To this end, ignition delay times were measured using an RCM and a ST at pressures from 10 bar to 40 bar, in the temperature range of 590 K to 1215 K. The data so measured have been used to validate a new kinetic model. In addition, we used laminar flow reactor experiments to measure first stage IDTs at atmospheric pressure. To improve the model predictions, the mechanism has also been validated against available literature data [18,23,24,27–29,31].

Overall, the model has been validated in the temperature range of 298–1850 K, at pressures of 1 to 60 bar, for highly diluted conditions and fuel/air mixtures at lean, stoichiometric, and rich conditions and represents a significant improvement in the understanding of DMM oxidation.

2. Experimental description

2.1. Shock tube (ST)

DMM ignition delay time measurements were performed in the shock tube at PCFC RWTH Aachen University. This facility has been described in detail by Zhang et al. [33]. The stainless steel ST is an assembly of a 3.0 m curved driver section and a 4.1 m driven section with an inner diameter of 63.5 mm. Both sections are separated by a double-diaphragm chamber. The shock wave is generated by bursting two pre-scored aluminum diaphragms. Mixtures of helium and compressed air were used as the driver gas. The velocity of the incident shock wave was measured using five pressure transducers (PCB 113B22) distributed over the last 1.0 m of the driven section before the closed end-wall of the ST. The ST has a heating system, capable of heating the facility from ambient temperature up to 150°C and the maximum working pressure is 500 bar. The recorded data (initial pressure, temperature, and incident shock velocities) are used as input parameters for an in-house code to determine the reflected shock conditions (pressure and temperature), which are based on the Shock and Detonation (SD) Toolbox [34] developed for Cantera [35]. An ignition delay time is defined as the time interval between the arrival of the incident shock at the end-wall and the near instantaneous auto-ignition pressure increase (Fig. 1), using the record of the pressure transducer closest to the end-wall (9 mm).

Due to boundary layer effects and non-ideal bursting of diaphragms, shock attenuation was identified [36]. For longer IDTs (above 1 ms), where these facility effects become important, an average pressure rise of $dp/dt = 7\%/ms$ was observed. Since this effect can shorten the resulting IDTs in the measurements [37], this pressure rise was converted into a corresponding effective volume-time profile and was included in our kinetic simulations.

The uncertainties in the measurement of the initial temperatures and pressures, the shock velocities and the thermodynamic data of the fuel/air mixture lead to an uncertainty in the temperatures and pressures behind the reflected shock of $\pm 0.7\%$ and $\pm 1.5\%$, respectively [33,38]. However, non-ideal effects, such as inhomogeneous ignition are difficult to quantify and can lead to an additional scatter of IDTs. Previous studies reported good agreement between different ST facilities within a range of $\pm 20\%$ [39]. This range is assigned as the uncertainty range of the measured IDTs in this study, where the observed scatter of the experiments is below this value.

Mixtures were prepared in a PTFE-coated stainless steel mixing vessel with a volume of 40 liter, and the shock tube with the attached mixing vessel including manifold were electrically heated to avoid condensation of fuels with a low vapor pressure. The initial temperature distribution is monitored by several type-T thermocouples distributed throughout the complete setup. During the DMM IDT measurements, heating to 40°C was sufficient to ensure no condensation for all experiments. The mixture composition was prepared by monitoring partial pressures using two static pressure transducers (STS ATM 1st 0–500 mbar, STS ATM 1st 0–10 bar).

2.2. Rapid compression machine (RCM)

The IDT measurements for DMM oxidation at low temperatures were conducted in the rapid compression machine (RCM) at PCFC RWTH University Aachen, and the facility has previously been described in detail [40,41]. This facility is a pneumatically driven, hydraulically stopped, single-piston machine. The diameter of the cylindrical combustion chamber is 50 mm and it has a 250 mm stroke. The piston head is designed with a crevice behind the piston face to minimize roll-up vortices, and an adjustable end-wall allows for the variation of the compression ratio of $\varepsilon = 9\text{--}32$. The RCM can operate up to 1000 bar peak pressure for reactive experiments. Each experimental condition is accompanied by a non-reactive experiment. In a non-reactive mixture the oxygen content of the corresponding reactive mixture is replaced by nitrogen so that the facility effects can be characterized, i.e. the effect of the compression phase and heat loss to the walls inducing a pressure decrease after the end-of-compression (EOC). The pressure histories of the non-reactive experiments for each individual condition are converted into an effective volume-time profile by assuming isentropic compression and expansion. These volume-time profiles are then prescribed in simulation as a boundary condition as described in detail in [42,43]. The non-reactive pressure traces are essential to accurately simulate the RCM experiments and are included as separate files in CHEMKIN format in the Supplementary material. The in-cylinder pressure is monitored using a KISTLER 6125C11U20 pressure transducer combined with a charge amplifier (KISTLER 5018). Figure 2 shows the definition of IDT, which is the time between EOC and the steep pressure rise due to the ignition event. The RCM is equipped with a heating system covering the reactor chamber with the attached manifold. The initial temperature in all experiments was set to 50°C and the system was controlled and monitored using several type-T thermocouples. The mixtures were prepared using two static pressure transducers (STS PTM 0–500 mbar, STS PTM 0–5 bar). Subsequently, a mixing time of at least 30 minutes was allowed in order to ensure homogeneous mixing for the ST and RCM experiments. Different

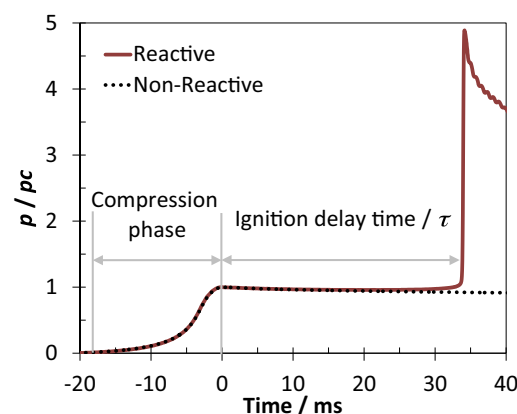


Fig. 2. Pressure trace showing ignition delay time of DMM. Experimental conditions: $p_c = 19.7$ bar, $T_c = 614$ K, $\phi = 1.0$ in air, $\tau = 34$ ms.

mixing times were tested with the result that shorter mixing times even as short as 15 minutes resulted in no recognizable difference in measured IDTs. In preparing the mixtures, the purities of the oxygen, nitrogen, and DMM used were at least 99.999%, 99.999% and 99.7%, respectively. All gases were obtained from Westfalen AG. Other fuel specifications are listed below. The stoichiometric fuel/air mixtures with an O_2/N_2 ratio of 21/79 contain 5% DMM. O_2 and N_2 were added to the mixing vessel after the initial vaporization of DMM.

Uncertainties in the EOC-conditions are estimated considering the uncertainties in (i) the initial pressure and temperature, (ii) in the mixture preparation, and (iii) the dynamic pressures [44]. These effects lead to an uncertainty of ± 5 K in the EOC-temperature. However, similar to the behavior in the shock tube, effects such as non-ideal ignition cannot be quantified but may lead to additional scatter in the IDT data. Again, similar to the ST measurements, previous studies reported good agreement between kinetic simulations and experimental results from different RCM facilities in case their respective facility effects are taken into account within a range of $\pm 20\%$ [39]. This range is assigned as uncertainty range of the measured IDTs in this study where the observed scatter of the experiments is below this value.

2.3. Laminar flow reactor (LFR)

The atmospheric pressure laminar flow reactor (LFR) at the Institute for Combustion Technology at RWTH Aachen University allows the measurement of first stage IDTs of up to 3500 ms at temperatures below 720 K. The experimental setup has been described in detail in [45] and only a brief description is provided here. The quartz tube LFR has a length of 4800 mm and is placed inside an external oven to achieve close to isothermal conditions across its length. The reactor is helically shaped with an incline of 9° and an outer helical diameter of 340 mm. The inner tube diameter is 20.6 mm with a wall thickness of 2.2 mm. The LFR is fitted with 48 type-N sheathed thermocouples along its length, each separated by an 89 mm distance. The thermocouples have a sheath diameter of 1 mm and their tips are positioned within a distance of 3 mm from the reactor wall to identify the ignition location with a minimal disturbance to the flow inside the LFR. The temperature change across the height of the flow reactor is minimized by circulating heated air around the reactor tubes in the oven.

High-purity DMM (Solvalid DMM >99.7%, INEOS Paraform GmbH & Co. KG), nitrogen (99.5%, Westfalen AG), and compressed dehydrated air were used for the LFR experiments. The fuel flow rate is controlled via a Brooks Quantim Coriolis mass flow controller (QMB3) with an accuracy of $\pm 0.2\%$ of the flow rate. The

fuel is stored under a nitrogen atmosphere in a stainless steel tank and is vaporized using an aDROP vaporizer operated with nitrogen carrier gas flow. The oxidizer (dehydrated air) and diluent (nitrogen) flow rates are controlled via Brooks mass flow controllers (SLA5850), which have an accuracy of $\pm 1\%$ of the set point for flow rates between 20–100% of the full scale and $\pm 0.2\%$ of the full scale rate for flows below 20% of full scale. The fuel and oxidizer streams are heated separately and are then rapidly mixed in a fuel injector to form a homogeneous mixture at the inlet of the LFR [45]. The fuel injector utilizes the design strategy of the flat flame burner of Zhang et al. [46], which forms a homogeneous mixture within a short distance of 1.5 mm [45,47]. Maximum temperature variations of ± 2.6 K were observed along the flow reactor length to the ignition location.

3. Chemical kinetic model development

In this study (Part II), a kinetic model has been developed to describe the chemical oxidation mechanism of DMM at engine-relevant conditions. The sub-mechanism for DMM consists of 185 new reactions and 37 species and is implemented using AramcoMech2.0 as the C₀–C₄ base mechanism [48]. The reaction rate constants for DMM are based on analogies to dimethyl ether (DME), diethyl ether (DEE), as well as *n*-pentane and are complemented with *ab initio* calculations whenever available, as described below.

For DME, the rate constants of the base mechanism from NUI Galway are used and originate from the work of Burke et al. [49]. Analogies with DEE and *n*-pentane were performed using the recent detailed kinetic mechanisms from Sakai et al. [50], and Yasunaga et al. [51] for DEE, and Bugler et al. [52] for *n*-pentane. The *ab initio* thermochemistry and kinetics described in Part I of this study are used in the present kinetic model [32]. Furthermore, relevant reaction rate coefficients of the DMM mechanism presented by He et al. [30] and Vermeire et al. [31] are implemented in the present kinetic model.

The following description will elucidate the details of the newly developed DMM mechanism. In general, the kinetic model development follows the reaction classes approach originally proposed by Curran et al. [53] and recently revised by Bugler et al. [54]. However, several classes are not listed here, due to the absence of C–C bonds in the molecular structure of OME_x. The model is validated against the IDT measurements from this study but also against IDTs [23,24,27], laminar burning velocities [27] and species concentrations [28,29,31] available in the literature. Furthermore, the ignition results of this model were compared to simulations of the published mechanism proposed by He et al. [30] and Vermeire et al. [31]. However, some of the validation targets are presented as Supplementary material Chapter 3. The newly developed model (chemical kinetics, thermodynamics and transport properties) is provided in CHEMKIN format and available as Supplementary material.

3.1. Thermochemical data

For the simulation of the oxidation of DMM, the thermodynamic data are essential. Enthalpies, entropies, and heat capacities over a range of temperatures, for several important species of DMM, e.g. fuel (RH), fuel radicals (primary C–O site: $\dot{R}1 = \text{CH}_3\text{OCH}_2\text{OCH}_2$ and secondary O–C–O site: $\dot{R}3 = \text{CH}_3\text{OCH}_2\text{OCH}_2$), fuel radical peroxides (RO_2), hydroperoxides (ROOH), and internal hydrogen abstraction products (QOOH), were taken from Part I of this study [32]. The structure of DMM is shown in Fig. 3. Part I of this study used high-level *ab initio* methods to derive thermochemical data (within 1 kcal/mol uncertainty) and fitted NASA polynomial expressions from 100 K to 1000 K and

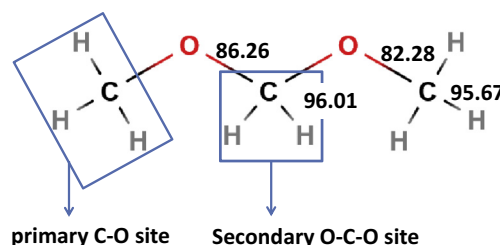


Fig. 3. Bond dissociation energies (BDEs) of dimethoxymethane. Unit kcal/mol.

Table 1

High-temperature reaction classes.

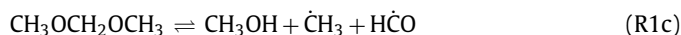
High-temperature reaction classes		
Class ID	Reaction class	Rate constants source
1	Unimolecular decomposition of the fuel	[49,51]
2	$\text{RH} \rightleftharpoons \dot{\text{R}} + \text{H}$	[58]
3	$\text{RH} + \dot{\text{X}} \rightleftharpoons \dot{\text{R}} + \text{XH}$	[30–32,49,51]
4	C–O: β -scission of $\dot{\text{R}}$ radicals	[32]
5	O–C–O: β -scission of $\dot{\text{R}}$ radicals	[32]
6	$\dot{\text{R}}$ radical isomerization	[32]

1000 K to 4000 K. The thermochemical data of the other relevant radicals and stable species were estimated with the Benson group increment theory implemented in THERM [55] or taken from the available literature [30,56]. An overview of all species, their naming and their source for the thermodynamic data is provided in the Supplementary material (Table S2).

3.2. Reaction rate coefficients

The major classes of elementary reactions are segmented into a high- and low-temperature application and are described in the following sections. In contrast to previous studies, pressure-dependent rate coefficients are assigned to a wide variety of reaction classes. The 6 high-temperature reaction classes considered in the development of this model are listed below in Table 1:

For DMM, four possible unimolecular decomposition reactions are considered:



For the unimolecular decomposition (C–O bond breaking) reactions, the rate constants were assigned by analogy with DME and DEE. Reaction R1a was adopted by analogy with DME decomposition ($\text{CH}_3\text{OCH}_3 \rightleftharpoons \dot{\text{C}}\text{H}_3 + \text{CH}_3\dot{\text{O}}$) in the base mechanism. The reaction resulting in the production of a methoxy and a methoxymethyl radicals was assigned by analogy with the decomposition of DEE producing ethoxy and ethyl radicals [51]. The rate parameters of the DEE unimolecular decomposition reaction with the products ethanol and ethene [51] were used to estimate the rate coefficients of a third possible decomposition reaction of DMM (reaction R1c). In the previous modeling study of Hu et al. [24], the unimolecular fuel decomposition reaction R1c was identified as a fuel-specific chain initiation reaction but was not included in the recent literature models of DMM [30,31]. The rate parameters of reaction R1d were taken from the DMM sub-mechanism of He et al. [30], which was originally derived from the DEE model of Yasunaga et al. [57].

Table 2

Low-temperature reaction classes.

Low-temperature reaction classes		
Class ID	Reaction class	Rate constants source
7	Addition of O ₂ to DMM radical ($\dot{R} + O_2 \rightleftharpoons \dot{R}O_2$)	[49]
8	$\dot{R} + R'\dot{O}_2 \rightleftharpoons \dot{R}O + R'\dot{O}$	[52]
9	Peroxy radical isomerization ($\dot{R}O_2 \rightleftharpoons \dot{Q}OOH$)	[31]
10	$\dot{R}O_2 + H\dot{O}_2 \rightleftharpoons ROOH + O_2$	[52]
11	$\dot{R}O_2 + H_2O_2 \rightleftharpoons ROOH + H\dot{O}_2$	[52]
12	$\dot{R}O_2 + CH_3\dot{O}_2 \rightleftharpoons \dot{R}O + CH_3\dot{O} + O_2$	[52]
13	$\dot{R}O_2 + \dot{R}O_2 \rightleftharpoons \dot{R}O + \dot{R}O + O_2$	[49]
14	$ROOH \rightleftharpoons \dot{R}O + \dot{O}H$	This study
15	$\dot{R}O$ decomposition	[49]
16	$\dot{Q}OOH \rightleftharpoons \dot{Q}O + \dot{O}H$ (cyclic ether formation)	[31]
17	$\dot{Q}OOH$ decomposition (β -scission products)	This study, [31]
18	Addition of O ₂ to $\dot{Q}OOH$ ($\dot{Q}OOH + O_2 \rightleftharpoons O_2\dot{Q}OOH$)	[49]
19	Isomerization of $\dot{O}_2\dot{Q}OOH$ and formation of ketohydroperoxide and $\dot{O}H$	[31]
20	Decomposition of ketohydroperoxide to form oxygenated radical species and $\dot{O}H$	This study
21	Cyclic ether reactions with $\dot{O}H$ and $H\dot{O}_2$	[30]

The dissociation of primary or secondary hydrogen atoms from DMM are conventionally only important at very high temperatures (>1800 K) and these have been neglected in previous works [30,31]. However, for a comprehensive description of the kinetics of DMM we have included this reaction class. Reaction parameters were estimated from rate coefficients of DME [58]. The unimolecular dissociation rate coefficients ($RH \rightleftharpoons \dot{R}1 + H$ and $RH \rightleftharpoons \dot{R}3 + H$) are related based on their bond dissociation energies (BDEs) [32].

Rate constants for the H-atom abstractions by O₂, $\dot{O}$, $H\dot{O}_2$, $CH_3\dot{O}$, CH_2OCHO , $CH_3O\dot{C}O$, $CH_3O\dot{C}H_2$, $H\dot{C}O$, and $\dot{C}H_2OH$ have been adopted from the work of Vermeire et al. [31], though only abstractions by O₂, $\dot{O}$, $H\dot{O}_2$ and $CH_3\dot{O}$ were calculated by quantum mechanics in that work. One of the most sensitive reactions for DMM IDTs is H-atom abstraction by $CH_3\dot{O}_2$ radicals. However, this reaction class has not yet been calculated in previous studies. Hence, we have derived rate coefficients for abstraction by $CH_3\dot{O}_2$ radicals based on those for H-atom abstraction by $H\dot{O}_2$ radicals [31]. The A-factor has been decreased by 20% which reflects the lower reactivity of $CH_3\dot{O}_2$ relative to $H\dot{O}_2$ radicals [49,59]. H-atom abstraction by methoxy radical was increased by a factor of two compared to the original rates recommended by Vermeire et al. [31], due to their influence at temperatures in the transition from low- to intermediate-temperature (700–950 K). The rate coefficients for H-atom abstraction by $\dot{O}H$ radicals were adopted from the DMM sub-mechanism by He et al. [30]. These rate coefficients were implemented in the PCFC DMM model because the trend of this calculation is similar to the $\dot{O}H$ radical measurements with DMM of Vovelle et al. [60]. However, the A-factor of the rate coefficient for the H-atom abstraction at the secondary site was decreased by approximately 25% in order to achieve better agreement with experimental data. This change is within the general uncertainty of the reported calculations. H-atom abstraction from DMM by radical $\dot{R}1$ resulting in the production of DMM and radical $\dot{R}3$ is adopted by using an analogy for the same type of abstraction reaction from DEE [51]. Furthermore, high-level quantum mechanics calculations from Part I [32] for H-atom abstractions by hydrogen ($\dot{H}$) and methyl ($\dot{C}H_3$) radicals for both primary and secondary radical sites of DMM are used. The H-atom abstraction reactions by other radical species, i.e. $\dot{R}O_2$, $\dot{O}_2\dot{C}HO$ and $\dot{O}CHO$ were assigned by analogy with DME from the base mechanism. All remaining reaction classes in Table 1 are included as proposed in Part I [32] and calculated using *ab initio* methods. Low-temperature reaction classes included in this study are listed below in Table 2.

In the low-temperature mechanism, the pressure-dependent rate coefficients for the reactions of $\dot{R} + O_2$ and $\dot{Q}OOH + O_2$ (Reaction classes 7 and 18) are adopted by analogy with DME. The rate

constants for the addition of $\dot{R}1$ to O₂ are assigned according to the corresponding reaction in the DME mechanism, while for the reaction involving the addition of $\dot{R}3$ to O₂, the rate coefficients are increased by a factor of three. In so doing, the relation of both reactions ($\dot{R}1 + O_2 \rightleftharpoons \dot{R}1\dot{O}_2$ and $\dot{R}3 + O_2 \rightleftharpoons \dot{R}3\dot{O}_2$) becomes similar to the ratio recommended for the addition of the primary and secondary site to O₂ in *n*-alkanes [52].

The values of the rate coefficients of reaction classes 8, 10, 11, and 12 are either lacking or are not calculated in the recent literature models of DMM [30,31] and are therefore assigned by analogy to *n*-pentane [52]. For DMM, three different reactions for internal isomerization ($\dot{R}O_2 \rightleftharpoons \dot{Q}OOH$) are possible (reaction class 9) involving a 6-membered transition state ring and an H-atom abstraction on the primary site ($\dot{Q}3OOH$) and on the secondary site ($\dot{Q}1OOH$) as well as an 8-membered transition state ring and a H-atom abstraction on the primary site ($\dot{Q}1fOOH$). For all three reactions, the *ab initio* rate coefficients of Vermeire et al. [31] have been adopted. These rate coefficients are similar to the calculation results of He et al. [30] and agree within a factor of 2 to 3 to each other. Reaction classes 13 and 15 ($\dot{R}O_2 + \dot{R}O_2 \rightleftharpoons \dot{R}O + \dot{R}O + O_2$ and $\dot{R}O$ decomposition) are deduced based on analogy with the DME sub-mechanism in the base mechanism. Reaction class 14 ($ROOH \rightleftharpoons \dot{R}O + \dot{O}H$) is estimated by analogy with reaction class 20 (decomposition of ketohydroperoxides), which will be discussed later. Possible various cyclic ether formations (reaction class 16) were calculated by Vermeire et al. [31] and are adopted in our model. However, the formation of methoxy-1,3-dioxetane (C3H6O3cy13) and $\dot{O}H$ radical from the species ($\dot{Q}OOH$) with the hydroperoxide group on the primary site was reduced by a factor of 2 in order to provide better agreement with the ignition delay measurements for the transition from low- to intermediate temperature. This agrees with the trend observed by comparing analogous reactions in DEE [50] and *n*-pentane [52]. The formation of the cyclic ethers 2,4-dimethyl-1,3-dioxetane and 2-ethyloxetane in the DEE and *n*-pentane mechanisms are significantly slower than the calculated reaction rates by Vermeire et al. [31] and He et al. [30]. IDT predictions at low and intermediate temperatures are very sensitive towards the two possible β -scission reactions of $\dot{Q}1OOH$. Thus, the dissociation kinetics of $\dot{Q}1OOH$ radicals are predicted based on geometries and harmonic frequencies at the B2PLYP-D3/6-311++G(d,p) level of theory, using single-point energies at the CCSD(T)/CBS(aug-cc-pVXZ; X=D, T) level of theory. Since an-harmonic contributions from internal torsional motions would mostly cancel for unimolecular dissociation reactions, all internal motions are treated as harmonic oscillators. Master Equation (ME) simulations for temperatures from 500 K to 2000 K and

Table 3

Modified Arrhenius parameters of the temperature- and pressure-dependent thermal β -scission rate constants of Q10OH radicals from DMM.

Pressure (atm)	A (cm ³ /mol/s)	n (-)	E _a (cal/mol)
$\text{CH}_3\text{OCHOCH}_2\text{O}_2\text{H} \Rightarrow \dot{\text{C}}\text{H}_3 + \text{HO}_2\text{CH}_2\text{OCHO}$			
0.01	8.245×10^{13}	-1.881	10,397.3
0.10	2.262×10^{17}	-2.579	12,802.0
1.00	8.880×10^{21}	-3.590	16,268.2
10.0	1.774×10^{26}	-4.525	20,206.8
100.0	5.624×10^{27}	-4.652	22,993.8
∞	6.835×10^{11}	0.429	17,979.4
$\text{CH}_3\text{OCHOCH}_2\text{O}_2\text{H} \Rightarrow \text{CH}_3\text{OCHO} + \dot{\text{C}}\text{H}_2\text{OOH}$			
0.01	5.470×10^{41}	-10.293	22,198.6
0.10	1.800×10^{45}	-10.976	24,819.9
1.00	1.190×10^{45}	-10.496	26,591.7
10.0	1.420×10^{41}	-8.900	27,134.0
100.0	5.020×10^{31}	-5.717	25,098.4
∞	2.540×10^{12}	0.374	18,486.1

pressures from 0.01 atm to the high-pressure limit are used to obtain rate coefficients (Table 3). Collisional energy transfer is modeled as $\langle \Delta E_{\text{down}} \rangle = 200 \text{ cm}^{-1} (T/300 \text{ K})^{0.85}$ and the collision rates are calculated according to the Lennard-Jones collision frequency, with the parameters $\varepsilon = 81.1 \text{ K}$, 351 K and $\sigma = 3.41 \text{ \AA}$, $6.65 \text{ \AA}$ for argon and Q10OH radicals, respectively. The parameters of Q10OH are taken from C₇H₁₆ as the closest analog in the study of Hippler et al. [61]. Rate constants are fitted with the modified Arrhenius equation (Table 3).

The thermal rate constants for β -scission of DMM radicals (cf. Kopp et al. [32]) and of Q10OH (cf. present work) are limited by the mixing of chemical and collisional eigenvalues of the ME at high temperatures [62]. In order to avoid the inadequate extrapolation of the modified Arrhenius equation beyond the highest accessible temperature T_{max} [63], we used the high-pressure limit rate constants to guide extrapolation: $k_{\text{extrap}}(T, p) = k(T_{\text{max}}, p) (dk_{\text{inf}}(T)/dk_{\text{inf}}(T_{\text{max}}))^m$, with m being fitted to reproduce the slope $dk(T_{\text{max}}, p)/dT$. This extrapolation scheme makes use of the temperature-independent availability of the high-pressure limit rate constants. One β -scission of the Q10OH radical occurs at the methoxy group with the products $\dot{\text{C}}\text{H}_3$ and hydroperoxy-methyl formate ($\text{HO}_2\text{CH}_2\text{OCHO}$). IDT predictions are very sensitive to the rate of this reaction at both high and low temperatures. A slight increase of the A-factor by a factor of 1.65 was necessary to improve agreement with the experimental ignition delay times. Vermeire et al. [31] also calculated the reaction rate coefficient of this reaction but their results present a higher rate coefficient over the entire temperature regime leading to a significant under-prediction of the reactivity using their rate constant. The other β -scission leads to the products methyl formate (MF, CH_3OCHO) and hydroperoxy-methyl radical ($\dot{\text{C}}\text{H}_2\text{OOH}$). The pressure-dependent rate constant of this reaction calculated here is within the uncertainty range of the calculations of Vermeire et al. [31] and He et al. [30]. However, previous studies considered this reaction as a concerted reaction directly forming methyl formate, formaldehyde, and a hydroxyl radical. The remaining β -scission reactions of reaction class 17 are adopted from the calculations of Vermeire et al. [31]. The formation of ketohydroperoxide (reaction class 19) has been assigned from the DMM mechanism of Vermeire et al. [31]. For the decomposition of the ROOH species (reaction class 14) and the ketohydroperoxide (reaction class 20), a rate coefficient of $k = 2.0 \times 10^{16} \times \exp(-21 \text{ 134 K}/T) \text{ cm}^3/\text{mol/s}$ has been assumed. This corresponds to the suggestion given by Burke et al. [39,64] with an increased activation energy by $1.5 \text{ kcal mol}^{-1}$ to give a good agreement with measured ignition delay times. The resulting rate coefficient is well within the uncertainty of the coefficients suggested by He et al. [30] and Vermeire et al. [31]. Reaction class 21 is adopted from the DMM sub-mechanism by

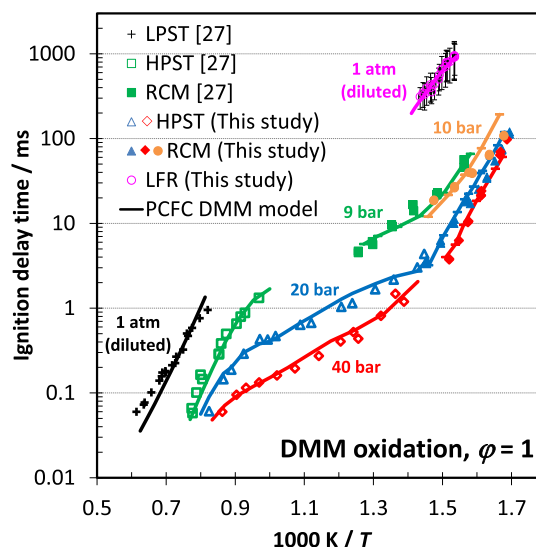


Fig. 4. Ignition delay times measured for stoichiometric mixture at pressures from 1 atm to 40 bar. Lines are simulations using PCFC DMM model including facility effects. High dilution for investigations at the LPST and LFR, HPST and RCM experiments in air. IDT measurements from the ST and RCM up to 9 bar reported by Gillespie [27], pressures > 9 bar and LFR: this study.

He et al. [30], which is based on the DME model by Rodriguez et al. [65]. All remaining reactions are again assigned from the DMM mechanism by Vermeire et al. [31]. Due to the absence of C–C bonds in DMM, the elimination reaction leading to the formation of $\text{H}\dot{\text{O}}_2$ radicals and an unsaturated molecule are not possible and consequently not considered in the kinetic mechanism. Furthermore, for the intermediate species dimethyl carbonate, the corresponding mechanism from Glaude et al. [56] was implemented in the PCFC DMM model. Finally, for methoxy methyl formate, a new sub-mechanism based on analogy of DMM and MF reactions was developed.

4. Results and discussion

4.1. Ignition delay times

To the best of the authors' knowledge the ignition delay time measurements in this study are the first at engine-relevant conditions (pressures 10–40 bar) using non-diluted fuel in air mixtures in ST and RCM (Fig. 4). In order to complement these data, results taken at highly diluted conditions at atmospheric pressure and in air at 9 bar [27] are also presented in the same plot.

IDTs longer than 3 ms are measured in the RCM and cover a temperature range between 590 K and 790 K. The range of short ignition delay times was covered using a ST thereby extending the temperature range to 1630 K. The experimental data is complemented with the corresponding simulation (including facility effects) in Fig. 4 that have been carried out using the LOGEresearch simulation software suite [66] and Cantera [35].

At high temperature, an Arrhenius type behavior is observed. Depending on pressure and temperature, e.g. for 20 bar at around 1100 K, a change in reactivity occurs leading to a lower increase in the IDT with decreasing temperature. At roughly 700 K, a second change in chemical reactivity can be observed resulting in a slightly stronger increase of the IDT with decreasing temperature. However, the slope of the IDT curve which represents a global activation energy is significantly less than that at high temperatures. This characteristic of the auto-ignition behavior of DMM is also visible for constant volume simulations at the same conditions and is not due to facility effects. In comparison to DME [49], DEE [50], or

longer alkanes [33,52], DMM shows no negative temperature coefficient (NTC) behavior, where IDTs increase with increasing temperature. Overall, higher pressures lead to a higher reactivity as is typical. At low temperatures, however, the reactivity for all pressures (9–40 bar) tends to converge. The kinetic mechanism developed in this study agrees well the data shown in Fig. 4. The experimentally observed changes of reactivity are predicted in the same way, and for the majority of data, the agreement between model and the measurements is within the uncertainty of the experiments. The RCM simulation, however, slightly over-predicts the ignition delay times at 10 bar and 20 bar at low temperatures. Nevertheless, the model can describe the ignition behavior of DMM in the high- and low-temperature regime.

In addition to the RCM and ST experiments, the first stage IDT measurements were performed in the LFR for stoichiometric DMM/O₂/N₂ mixtures at atmospheric pressure and temperatures between 651 K to 697 K under diluted conditions (O₂/(O₂ + N₂) = 15%). First stage ignition is identified by the axial position of the thermocouple reporting the highest temperature and is subsequently converted to a derived 0-D first stage IDT. Details regarding the data analysis of the LFR are discussed in [67]. Over the temperature range measured, the kinetic model is able to predict the first stage IDTs with good accuracy.

Further validation of the PCFC DMM model against IDT measurements for fuel-lean and fuel-rich conditions [23,24,27] are shown as Supplementary material (Fig. S1–S8). Overall the comparison between these data and the model predictions show good agreement. However, the results of the simulation do not agree to the data of Zhang et al. [23] within the experimental uncertainty, despite the fact that the model is capable of predicting the data of Gillespie [27] and Hu et al. [24] at similar conditions. Previous investigations have shown that the IDT measurement depends on the location of the detector. In the experiments of Zhang et al. [23], ignition was monitored using sidewall pressure and light emission measurements. In contrast, Hu et al. [24] carried out the measurements using a detector positioned at the end-wall. Therefore, both investigations have different definitions of the ignition delay time. Further effects of the difference in these experimental methods are described in detail in the work of Petersen [68]. In summary, sidewall measurements result in a shorter ignition delay times compared to ignition delay time measurements deduced from an end-wall port. The discrepancy of this sidewall pressure measurements increases with increasing distance from the end-wall and with increasing temperature. Furthermore, in the work of Zhang et al. [23] the non-ideal boundary layer effect is neglected, but in the IDT study of Hu et al. [24] information regarding the shock attenuation is given, and this has been taken into account in the simulation.

For comparison, the performance of the previously published model of He et al. [30] and Vermeire et al. [31] have been tested against the newly available data, although these models have not been validated for the conditions considered here. The prediction of the kinetic mechanisms from He et al. [30] shows good agreement between the experimental data and kinetic model predictions of this study in the low- and high-temperature range. However, a lower reactivity is predicted in the temperature range 700 K < T < 1000 K. The simulation results of the Vermeire et al. model [31] also under-predict the reactivity of DMM over the entire range of temperatures investigated. A representative plot for this comparison at 20 bar and stoichiometric conditions can be found as Supplementary material (Fig. S9).

4.2. Reaction pathway and sensitivity analysis

To further illustrate the oxidation scheme of DMM, pathway analyses were performed at different temperatures (between 600 K

and 1150 K) and at a pressure of 20 bar at stoichiometric conditions in air (Fig. 5). The calculations are carried out at the time of 20% fuel consumption. The main consumption of DMM occurs via H-atom abstraction by hydroxyl (OH) radicals over the entire temperature range, resulting in the formation of the two possible fuel radicals, R1 and R3. Similar to the results of other previous studies [23,29], the favored abstraction site is the primary one. Apart from H-atom abstraction by OH radicals, the fuel is consumed via H-atom abstraction by H, HO₂, CH₃, CH₃O, and CH₃O₂ depending on the temperature, but each of these contributions is well below 10%. The fuel radicals are consumed by β -scission or reactions with molecular oxygen to form RO₂ radicals. An important result of the pathway analysis is that the secondary DMM radical dominantly decomposes to MF and CH₃ radicals almost over the entire temperature range. This is due to the low barrier height of the R3 β -scission reaction cleaving the weak C–O bond (Fig. 3). The corresponding bond dissociation energy is significantly lower compared to typical C–C bond strengths in alkanes but is also weaker than the other C–O bond which is cleaved in the R1 β -scission reaction. Consequently, almost no peroxy radicals are formed via this channel inhibiting the alkane-like low-temperature chemistry pathways towards chain branching. Only at the lowest temperatures (600 K) are significant amounts of R3O₂ peroxy radical formed which can undergo typical low-temperature reactions.

In contrast to this, R1 adds to O₂ in appreciable amounts and this is its primary consumption pathway competing with the β -scission reaction with increasing temperature. After addition to O₂, two possible isomerization reactions consume the R1O₂ peroxy radical, whereby the isomerization reaction involving a six-membered transition state and H-atom abstraction at the secondary site dominates over the entire temperature regime. In the subsequent reactions, the hydroperoxide radical Q10OH preferentially decomposes to a cyclic ether species (reaction class 16) at intermediate and high temperatures. The second addition to O₂ becomes dominant only at low temperatures. The second hydroperoxide radical Q1fOOH of the primary consumption pathway is formed via hydrogen atom migration from the primary site to the peroxy group through an 8-membered TS ring and mainly produces 1,3,5-trioxane and a hydroxyl radical at low temperatures. The second addition to O₂ only becomes dominant at low temperatures, similar to Q10OH.

In addition to the reaction pathway analysis, a sensitivity analysis was also performed for stoichiometric DMM/air at p = 20 bar and temperatures of 600 K, 825 K, and 1150 K. To test for sensitivity of IDT (τ) predictions, the rate constant of each reaction was increased (τ_+) and decreased (τ_-) by a factor of two. The ratio of the change in the IDT, $S = \log_{10}(\tau_+/\tau_-)/\log_{10}(2.0/0.5)$, is taken as the sensitivity of IDT towards that particular reaction [49]. A positive sensitivity coefficient implies that the IDT increases with an increasing rate constant of the particular reaction and vice versa for a negative sensitivity coefficient. Figure 6 displays the results of the sensitivity analysis highlighting the most sensitive reaction coefficients. Besides the decomposition of hydrogen peroxide, generally an important reaction for hydrocarbon fuels, C₁ reactions are important at high temperatures due to the high concentrations of methyl radicals that are produced by β -scission of the R3 fuel radical. For fuel specific reactions, the IDT is very sensitive with respect to the H-atom abstraction reactions by hydroxyl radicals. As discussed above, H-atom abstraction on the primary site can eventually lead to low-temperature chain branching reactions and consequently lead to a negative sensitivity coefficient. In contrast, H-atom abstraction on the secondary site is dominated by β -scission of the fuel radical leading to a suppression of the reactivity and positive sensitivities, respectively. The methyl radicals formed by this β -scission can subsequently add to molecular oxygen producing methyl-peroxy radicals resulting in a high sensitivity of

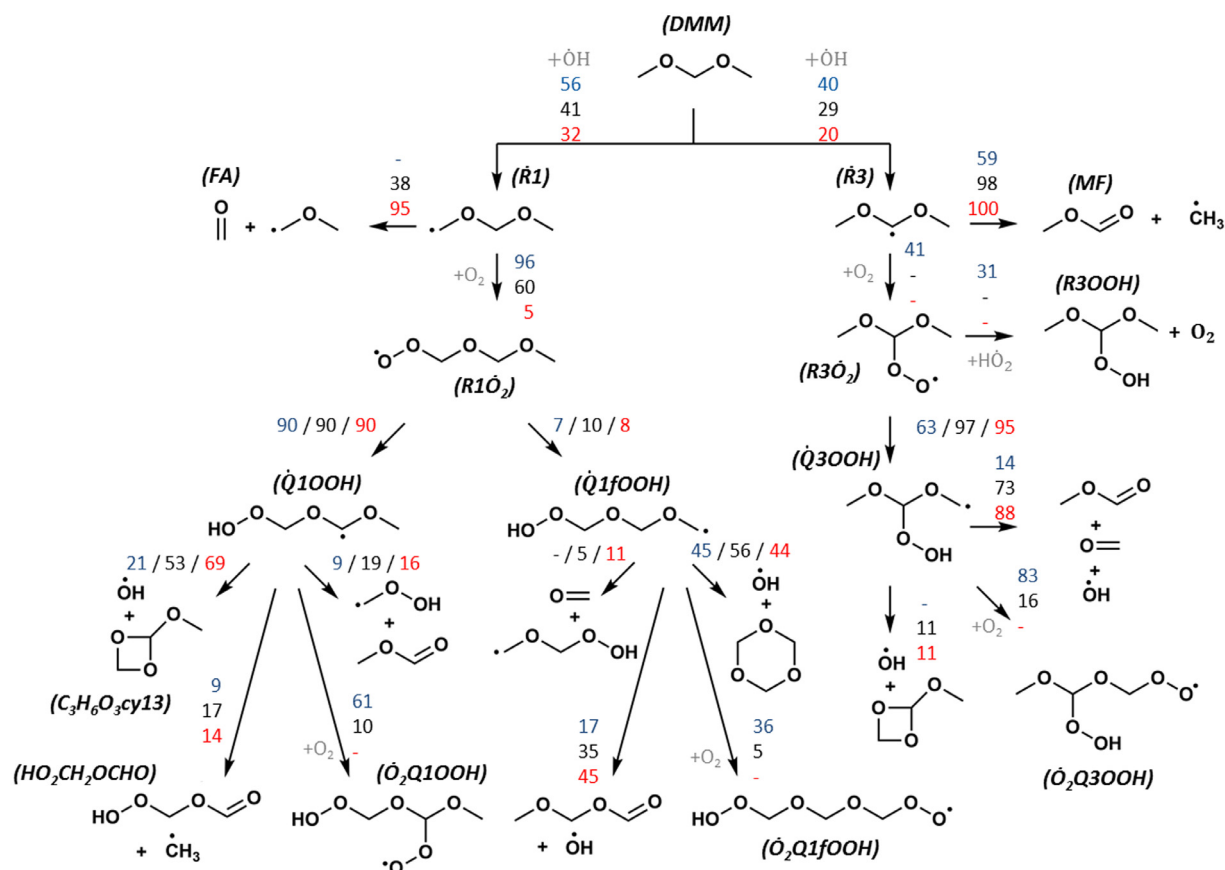


Fig. 5. The oxidation mechanism predicted by the model for temperatures of $T=600$ K (blue), $T=825$ K (black) and $T=1150$ K (red) at $p=20$ bar under stoichiometry conditions in air. The reaction flux analyses were performed when 20% of the initial fuel concentration was consumed. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

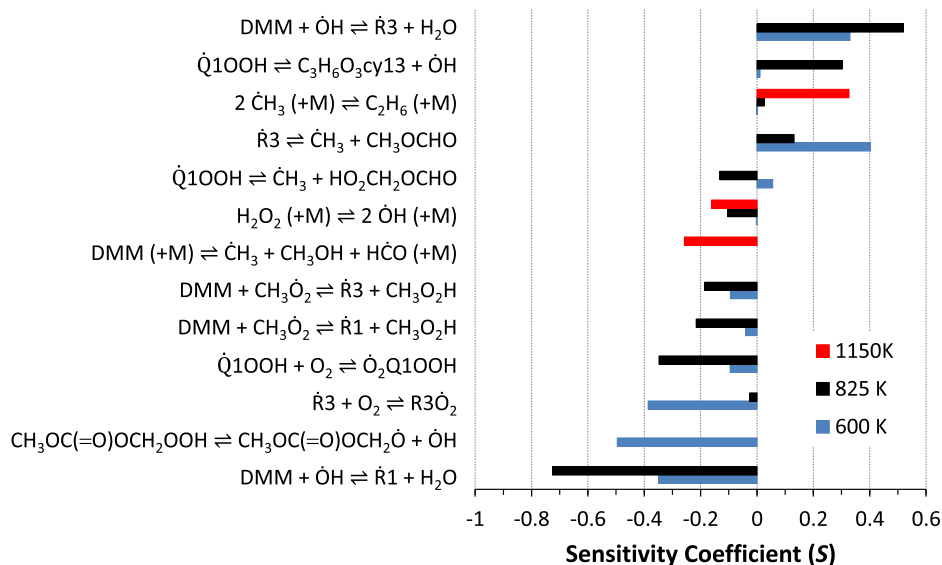


Fig. 6. Sensitivity analyses on IDTs for a range of temperatures under stoichiometric DMM in air conditions at 20 bar, $T=600$ K, 825 K and 1150 K. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

H-atom abstractions involving this radical. Especially at intermediate temperatures, H-atom abstraction by methyl-peroxy radicals strongly promotes reactivity. In this case there is little difference in sensitivity between the primary and secondary site, since the reactivity promoting effect results from the subsequent decomposition of the methyl hydroperoxide into a methoxy and a hydroxyl

radical. The formation of the cyclic ether methoxy-1,3-dioxetane via the $\dot{Q}100H$ radical inhibits reactivity at intermediate temperatures since it competes with the low-temperature pathways leading to chain branching. Similarly, at 600 K the dissociation of $\dot{Q}100H$ radicals ($CH_3OCHOCH_2O_2H \rightleftharpoons \dot{CH}_3 + HO_2CH_2OCHO$) inhibits reactivity due to its competition with the chain branching

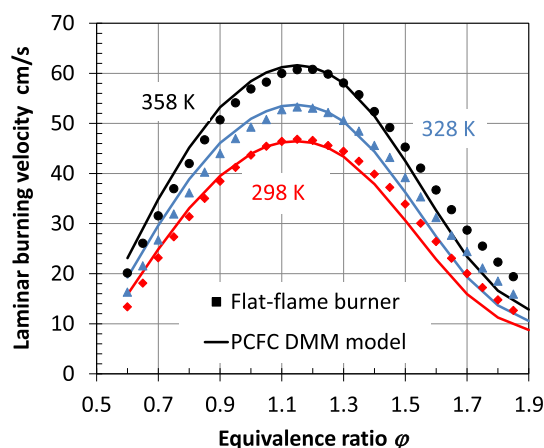


Fig. 7. Experimental data (symbols) reported by Gillespie [27] and kinetic model predictions (lines) for laminar burning velocities at equivalence ratios from 0.6 to 1.85 for initial temperatures of 298 K, 328 K and 358 K at atmospheric pressure in air.

pathway, but a flip in sensitivity coefficient with increasing temperature is observed. At higher temperatures, the second addition to O_2 and its subsequent chain branching pathways become less important (Fig. 5) and the major competition is between cyclic ether formation and β -scission of the Q10OH radical. The β -scission reaction forms a methyl radical and hydroperoxymethyl formate associated with DME low-temperature kinetics. This hydroperoxy-methyl formate can easily decompose and produces another hydroxyl radical. Altogether, the β -scission pathway can form both a methyl radical and a hydroxyl radical, whereas the formation of a cyclic ether produces only a hydroxyl radical. For this reason the β -scission pathway becomes reactivity enhancing at higher temperatures. Finally, the unimolecular decomposition of DMM forming methanol, methyl, and formyl radicals appears as the most sensitive reaction at high temperatures since it acts as a fuel-specific initiation reaction due to the formation of reactive radicals.

4.3. Laminar burning velocities

For further validation of the kinetic model, measurements of laminar burning velocities by Gillespie [27] were used for comparison of the new kinetic model predictions. The flat-flame burner facility at the Université de Lorraine used in the study by Gillespie has been described in detail previously [69–71]. The experiments were carried out at a pressure of 1 atm and at several initial temperatures of 298 K, 328 K, and 358 K and for equivalence ratios ranging from 0.60 to 1.85 in air. The experimentally obtained laminar burning velocities of Gillespie and the prediction of the developed kinetic model are shown in Fig. 7. Simulations of the flat-flame burner were performed using the freely propagating flame module within LOGEresearch [66], and included thermo-diffusive effects. A maximum number of grid points of 301 and maximum number of grid points per unit curvature and per unit gradient of 1.0 and 1.5 have been used for simulation. Grid convergence has been checked by increasing the maximum number of grid points per unit curvature and unit gradient together with reducing the relative tolerance limit by each a factor of two. The results indicate a change of the simulated laminar burning velocities of less than 1%. The maximum burning velocities of 46.8 cm/s, 53.3 cm/s, and 60.8 cm/s are observed at an equivalence ratio of $\phi = 1.15$ –1.2 for $T = 298$ K, 328 K, and 358 K. It is worth noting that the maximum burning velocities occur at a slightly richer mixture compared to paraffinic fuels for which a maximum is typically found

between $1.0 \leq \phi \leq 1.1$. The predictions of the kinetic model reach maximum burning velocities at an equivalence ratio of 1.15 for all initial temperatures. For an initial temperature of 358 K, the maximum laminar burning velocity shows a slight deviation between experimental measurement and model prediction. Furthermore, the simulation slightly over-predicts the experimental data on the fuel-lean side, while for fuel-rich mixtures, the prediction is about 3 cm/s slower than the experimental burning velocity values. Nevertheless, overall the laminar burning velocity simulation predicts the experimental trends well. Sensitivity analysis at atmospheric pressure and 328 K indicate typical reactions of the C_0 – C_1 base mechanism ($H + O_2 \rightleftharpoons \dot{O} + \dot{O}H$, $CO + \dot{O}H \rightleftharpoons CO_2 + H$, $HCO + M \rightleftharpoons CO + H + M$) as most important reactions for the laminar burning velocity as it is the case for most of hydro-carbon fuels. The burning velocity is not highly sensitive to fuel specific reactions. Indeed the most sensitive fuel specific reactions exhibit a sensitivity coefficient that amounts to only 3% compared to the most sensitive reaction in the entire system. The most sensitive fuel specific reactions belong to the high-temperature reaction classes, i.e. unimolecular decomposition of DMM forming an H-atom and a fuel radical, H-atom abstraction by $\dot{H}$ and $\dot{O}H$, and the β -scission of the fuel radicals.

4.4. Species measurements in flow reactors

Species measurements during the oxidation of fuels are highly valuable validation targets since they highlight details of the reaction pathways which cannot be observed by targeting global combustion parameters, such as IDTs or laminar burning velocities. In this regard, Vermeire et al. [31] presented data measured in a jet-stirred reactor for different equivalence ratios, at a pressure of 1.07 bar and a residence time of 2.83 s. Further, Daly et al. [18] reported data at 5.07 bar taken in a JSR. Finally, Marrodán et al. [28,29] reported investigations in a plug flow reactor under a wide range of temperatures (373–1373 K), pressures (1–60 bar), and air excess ratios ranging from 0 to 35. Representatively, Fig. 8 highlights a comparison of the simulation results and the experimental data from Vermeire et al. [31] for a stoichiometric DMM/oxygen/helium mixture showing the evolution of the major species namely, DMM, O_2 , CO, CO_2 , CH_4 , CH_2O , and CH_3OCHO . Simulations have been performed with the perfectly stirred reactor module in LOGEresearch [66]. While the overall prediction using the PCFC DMM model is satisfying, fuel consumption is slightly over-predicted at intermediate temperatures and as a consequence the mole fractions of some intermediates are over-predicted, particularly formaldehyde and methyl formate. In this intermediate temperature regime, FA and MF are formed by the β -scission reactions of $\dot{R}1$ and $\dot{R}3$ radicals of DMM, respectively. In addition, FA is also produced in significant amounts through the DME sub-mechanism, whereby DME radicals are produced by the β -scission reactions of the DMM radical $\dot{R}1$. Furthermore, the mole fraction of methanol ($MeOH$, CH_3OH) is clearly under-predicted by the model in the entire temperature regime investigated (Fig. S10). At low temperatures, $MeOH$ is formed through H-atom abstraction from DMM by methoxy radicals. At high temperatures, $MeOH$ is mainly produced from the unimolecular decomposition of DMM fuel by the following reaction: $CH_3OCH_2OCH_3 \rightleftharpoons CH_3 + CH_3OH + HCO$.

In this temperature regime (>850 K), the kinetic model shows good agreement with the experimental data indicating a good representation of the branching ratio between the two fuel radicals. The profiles of CO and CO_2 are well-predicted at low-temperatures, but above $T > 1000$ K, the discrepancy of the prediction is about 25 K. The methane profile is also well predicted despite a slight over-prediction overall, but the formation and consumption is mainly described in the base mechanism. Further validation comparing to species measurements is provided as Supplementary

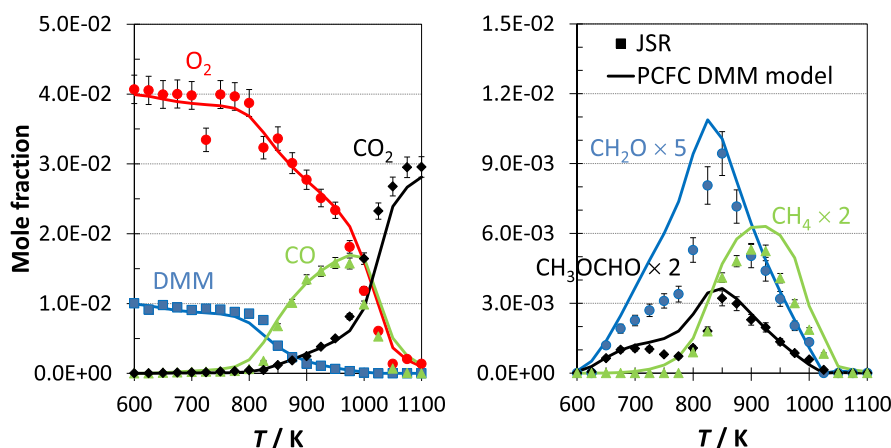


Fig. 8. Species mole fractions as a function of temperature for the oxidation of DMM. Comparison between experimental data (symbols) reported by Vermeire et al. [31] and simulation results (lines) for a DMM/oxygen/helium mixture at $p = 1.07$ bar, $\varphi = 1.0$ and 2.83 s residence time. In case of scaling for visibility both, experimental data and simulation results are scaled by the same factor.

material (Fig. S10–S15). Overall reasonable agreement between model and experimental result is achieved by the PCFC DMM model although some results indicate that there is room for optimization of the model. In order to improve the mechanism in this regard, the sub-chemistry of the intermediate species may need to be revised in detail which is out of the scope of this study.

5. Conclusions

This study presents a comprehensive investigation of the oxidation kinetics of DMM. Firstly, new measurements of IDTs at engine-relevant conditions were performed in a ST and an RCM. In addition, first stage IDTs measured in a laminar flow reactor at low temperatures are presented. Overall, new experiments have been performed for a wide regime of pressures (1–40 bar) and temperatures (590–1215 K).

The kinetic model development is based on the C_0 – C_4 AramcoMech2.0 and the state-of-the-art reaction rate class approach including information from recent works on dimethoxymethane. The resulting mechanism includes high- and low-temperature reaction pathways and has been tested against the experimental data from this study, but also against literature data, i.e. laminar burning velocities, various ST studies, and species measurements in jet-stirred and plug flow reactors.

Reaction pathway and sensitivity analyses revealed important reactions in the oxidation kinetics. They reveal that the low temperature reactivity of dimethoxymethane is largely controlled by the initial branching ratio for the production of the primary and secondary fuel radicals. For the primary fuel radical, conventional alkane-like low temperature pathways are possible, eventually leading to chain branching. Consumption of the secondary fuel radical, however, is controlled by fast β -scission over a wide temperature range inhibiting the pathway towards low temperature chain branching.

To summarize, the newly developed kinetic model is capable of predicting the global combustion parameters, i.e. IDT and laminar burning velocity, under various conditions well, mostly within the experimental uncertainty. Also species measurements in flow reactors can be predicted reasonably well. However, it is noted that there are some deviations in the prediction of intermediate species concentrations, e.g. for methyl formate. In this regard, a revision of the respective sub-mechanisms of these intermediates might be necessary in order to improve the prediction quality of the DMM model.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2018.12.026.

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